

WHEN DOES LEAVE-ONE-OUT SUFFICE? AUDITING MINIMAL-CAUSE ATTRIBUTION IN RAG

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ABSTRACT

RAG attribution is often scored per document although a response may have several alternative minimal causes. We formalize attribution as enumeration of all inclusion-minimal sufficient document sets and prove a boundary for grand-coalition leave-one-out (LOO): under monotone replay, LOO recovers the unique cause but returns only the intersection of alternatives. We introduce fully oracled unique/alternative/non-monotone (U/A/N) strata and audit black-box procedures on HotpotQA and Natural Questions with two 7B generators. Mask counts overstate cost when masks render to identical prompts; correct prompt-level caching removes an apparent advantage of a RAG-specific adaptation over standard enumeration. An enumerate-first prompt-uniform residual audit preserves perfect U/A recovery and raises source-clustered raw-N violation recall from .388 to .952, but a no-collapse stress test falls to .814. We prove uniform sampling minimax for an adversarial static hidden-witness subproblem. A frozen geometry-aware ordering subsequently fails its prespecified holdout gate. Thus a finite unviolated trace is diagnostic, not a monotonicity certificate.

Index Terms— retrieval-augmented generation, context attribution, sufficient causes, leave-one-out, evaluation audit

1. INTRODUCTION

RAG combines parametric generation with retrieved evidence [1, 2, 3], but attribution still asks which passages caused a fixed answer. Existing methods assign document scores or citations [4, 5, 6], but a scalar ranking does not identify a *set* of alternative minimal sufficient causes. This distinction matters in poisoning and forensic analysis: removing one path can leave another intact. Minimal sufficient subsets are established in explanation research [7, 8, 9]; our question is narrower: what can exact black-box replay establish for a fixed RAG context?

Contributions. We make four contributions. (1) *Identifiability*: we prove that LOO exactly recovers a unique monotone cause but only the common backbone of alternatives, and that an unviolated finite trace is not a monotonicity certificate. (2) *Benchmarking*: we construct fully enumerated U/A/N strata whose labels come from the realized generator truth table, never the planted template. (3) *Methodology*: we introduce an enumerate-first pipeline that protects all-cause search, deduplicates byte-identical rendered prompts, and spends only the residual physical-call budget on safety auditing; uniform class sampling is minimax in a static hidden-witness

game. (4) *Audit evidence*: corrected accounting removes an apparent method advantage; transfer, no-collapse stress, and two frozen holdout attempts expose where the diagnostic succeeds and where it must abstain.

Relation to prior work. Sufficient-input subsets and rationale benchmarks formalize local evidence [7, 10], while delta debugging minimizes a failure-inducing input [11]. ContextCite fits a KernelSHAP surrogate over context ablations [4, 12]; MIRAGE and SelfCite instead use model internals or citation supervision [5, 6]. Our all-cause endpoint differs from feature ranking and single-subset minimization [13, 14, 15]. Explanation quality also depends on the claim and evaluation target [16, 17]; we therefore make faithfulness assumptions explicit and prefer intervention-backed tests [18, 19, 20, 21, 22, 23, 24]. Post-hoc local surrogates can also be adversarially misleading [25]. Its standard baseline follows the same minimal/maximal duality used by MUS/MCS enumeration [26]. Property testing can reject functions that are far from monotone [27]; our trace proposition is narrower and makes no tolerant- testing lower-bound claim.

Data availability. Code, benchmark manifests, and replay artifacts are publicly available at <https://github.com/violetzxd/CoalitionTrace>.

2. PROBLEM SETUP AND BOUNDARIES

For documents $D = \{1, \dots, n\}$, let $r(S)$ be the complete rendered chat prompt (including the system message and generation marker) obtained by keeping the observed version of documents in S and their contrastive replacement otherwise. Deterministic replay defines $v(S) = \mathbb{K}[\text{NEM}(g(r(S))) = \text{NEM}(y_t)]$, where NEM is a frozen normalization and exact-match rule: a format-only extractor removes an “Answer:” prefix, later lines, and a trailing parenthetical before standard lowercase/article/punctuation normalization. A set C is an inclusion-minimal sufficient cause (relative to the fixed model, decoder, replacement, and target) when $v(C) = 1$ and $v(C') = 0$ for every $C' \subsetneq C$. Let $\mathcal{M}(v)$ contain all such causes. We evaluate exact cause-set match (ACEM), cause-set precision/recall/F1, observed non-monotonicity, and replays.

Proposition 1 (LOO boundary). Write $D_{-i} = D \setminus \{i\}$. If v is monotone and $v(D) = 1$, grand-coalition LOO returns

$$\{i : v(D_{-i}) = 0\} = \bigcap_{C \in \mathcal{M}(v)} C. \quad (1)$$

Hence LOO returns an inclusion-minimal sufficient cause exactly in the unique regime, $|\mathcal{M}(v)| = 1$. With alternatives it returns only

their common backbone. The proof follows because every sufficient set contains a minimal cause: deleting i fails exactly when every minimal cause contains i .

Corollary (one-query ambiguity test). Let B be the LOO set above. Under the same assumptions, $v(B) = 1$ iff $|\mathcal{M}(v)| = 1$. Thus at most $n+2$ replays (full, n deletions, and B) classify U versus A and recover the U cause. If B is sufficient it contains a minimal cause, which must equal the intersection and hence every minimal cause; the converse follows from Proposition 1. This certificate is invalid without monotonicity.

This motivates three oracle strata, all requiring $v(\emptyset) = 0$ and $v(D) = 1$: U has monotone v and one minimal cause; A has monotone v and at least two; N contains a Boolean cover-edge violation.

Proposition 2 (trace non-certifiability). Assume a partial trace over exact rendered-prompt classes has a monotone completion. If an entirely unqueried class represented by x has either a queried insufficient strict superset or a queried sufficient strict subset, then a non-monotone completion produces the identical trace: assign every mask in that whole unqueried class the opposite of the value forced by monotonicity. Thus absence of an observed violation cannot certify monotonicity for such a trace.

Proposition 3 (static hidden-witness minimax). After a fixed prefix, suppose M classes remain, exactly K are immediate witnesses, every non-witness returns one common miss, misses cannot create new witnesses, and all K -subsets are admissible. Any randomized adaptive audit of $b \leq M$ classes has worst-case detection at most $V = 1 - \binom{M-K}{b} / \binom{M}{b}$; uniform sampling without replacement attains V for every witness set. Indeed, under a uniformly hidden K -set, any fixed all-miss query set of size at most b intersects it with probability at most V , so averaging yields one instance no easier than V ; uniform sampling has that probability pointwise. In a real sequential audit, dynamic witnesses can only make the fixed-prefix expression a lower bound, not a global minimax claim.

3. METHODOLOGY

Figure 1 summarizes our five-stage procedure. The separation is deliberate: cause recovery, violation detection, and certification are different objectives. Safety queries never preempt the task-matched cause enumerator. The procedure maintains three persistent objects: a prompt-class cache H , a set $\widehat{\mathcal{M}}$ of verified minimal causes, and an exclusion frontier F . Every returned cause is replay-checked for sufficiency and member necessity; every safety decision is tied to an explicit comparable pair.

Stage 1–2: validate, cache, and diagnose. We first replay $\emptyset$ and D ; cases with $v(\emptyset) = 1$ or $v(D) = 0$ are out of scope. We then query every D_{-i} , insert the resulting prompt classes into H , and form backbone B . Under a justified monotonicity assumption, $v(B)$ implements the corollary’s U/A test; without that assumption it is only a diagnostic fast path.

Stage 3: protected all-cause enumeration. At exclusion E , query the maximal allowed set $D \setminus E$; prune if it is insufficient, otherwise minimize it by deterministic single-document deletions to C , verify C , add it to $\widehat{\mathcal{M}}$, and branch on $E \cup \{i\}$ for every $i \in C$. Duplicate exclusions are removed and every replay first consults H . Branching preserves a frontier entry disjoint from each undiscovered cause. Hence frontier exhaustion is conditionally complete under monotonicity; budget exhaustion returns discovered causes but abstains from completeness. This is a standard minimal-explanation enumerator [8, 9], not a new generic enumeration result.

Stage 4–5: residual audit and evidence grading. The post-

hoc wrapper runs the enumerator unchanged, then samples previously unseen rendered-prompt classes uniformly without replacement until the residual physical-call budget is spent. After each replay, its value is propagated to every byte-identical mask and compared with the observed subset closure. The audit stops at the first pair $L \subset U$ with $v(L) = 1, v(U) = 0$. The grader then emits one of four states: *exact-monotone* (frontier and relevant closure exhausted), *conditional-complete* (frontier exhausted under monotonicity), *violated* (a concrete reversal exists), or *budget-limited*. The record bundles causes, witness, logical masks, unique prompts, tokens, stopping reason, and cache multiplicities.

Why the composition helps. First, recovery is protected: the audit is strictly residual, so adding it cannot lower U/A ACEM at the same total budget. Second, exact prompt quotienting is cost-faithful and lossless for deterministic replay, unlike semantic merging. Third, positive and negative conclusions are intentionally asymmetric: one reversal is a sound refutation, whereas a clean partial trace triggers abstention rather than a false certificate. Finally, the output is auditable end to end—a reviewer can reconstruct each cause, cache hit, cost, and safety label from the trace instead of trusting a scalar score.

3.1. Exact replay accounting

Define $S \sim T$ iff $r(S) = r(T)$ byte-for-byte. This is the conservative exact request equivalence used by our cache, with model, decoder, scorer, and fixed randomness held constant; token-ID equivalence could merge additional byte-distinct prompts and is not claimed here. **Proposition 4 (quotient accounting).** Any adaptive policy depending only on its logical mask/response transcript can be simulated exactly with one call per distinct class in $\mathcal{Q} = 2^D / \sim$. Under b uniform mask draws, class q is included with probability $1 - \binom{2^n - m_q}{b} / \binom{2^n}{b}$, biased by its multiplicity m_q . For a *static* single witness class, when its identity does not change and every miss yields the same all-miss transcript, no adaptive policy using c distinct class calls guarantees detection above $c/|\mathcal{Q}|$: inclusion marginals sum to c , and class-uniform sampling attains the bound. Cache induction, avoidance counting, and averaging prove the claims. The quotient is lossless for calls, not a cause lattice, so queried values are propagated to all member masks before subset comparisons.

4. EVALUATION

We ask whether LOO exhibits the predicted U/A boundary; whether prompt-level accounting changes method comparisons; whether residual auditing improves N -witness recall without sacrificing cause recovery; and whether development gains survive transfer, no-collapse stress, and frozen holdouts.

4.1. Benchmark and realized ground truth

We start from public PoisonedRAG target questions [28] for HotpotQA and Natural Questions [29, 30]. Each $n = 8$ context contains contrastive document slots that instantiate one path (U), two disjoint or overlapping paths (A), or a suppressor/restorer pattern (N). We exhaust all 2^8 masks with greedy decoding. We retain a source only when its realized table satisfies the exact stratum definition; all failures and the screening funnel are retained in a hashed artifact provided in the submitted supplementary material. A source query, not a template or mask, is the sampling unit. The *primary registry* frozen test split contains 70 sources per dataset; later news, all-active, and

Enumerate first; audit only with the residual physical-call budget

1. Validate Replay empty + full invalid: out of scope	2. LOO test Delete each document form backbone B	3. Cause search Protect enumeration exhaust its frontier	4. Safety audit Sample unseen classes stop at a reversal	5. Grade output Causes + witness + cost certify or abstain
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A found reversal rejects monotonicity; an unviolated non-exhaustive trace remains inconclusive.

Fig. 1. CoalitionTrace methodology. Endpoint checks establish that the local question is well posed. LOO gives a cheap backbone diagnostic; the exclusion frontier searches all causes; residual unseen prompt classes are then audited. Only an observed reversal rejects monotonicity—otherwise a non-exhaustive trace abstains.

holdout audits use separately reported pools. Models are Qwen2.5-7B-Instruct and Mistral-7B-Instruct-v0.3 at pinned revisions.

4.2. Baselines, metrics, and protocol

Methods are LOO; hierarchical deletion; ten-seed repeated greedy deletion; ten-seed random masks; constrained KernelSHAP as a ContextCite-style surrogate; a standard monotone exclusion enumerator; and a RAG adaptation that adds an LOO backbone fast path. The post-hoc residual audit is evaluated separately from the frozen seven-method comparison, with ten seeds. All share the same SHA-256 prompt cache. The primary small- n budget is 48 unique prompts. Exact tables supply truth, not a deployable method. We bootstrap source queries (10,000 resamples); stochastic seeds are averaged within source and are never treated as samples. This intervention-based design tests faithfulness directly rather than treating attention or plausibility as a causal explanation [19, 18]; it also differs from training-data influence analysis [31].

Oracle and construction validity. Every stored table is checked for all 2^n masks, target normalization, $v(\emptyset)$, and $v(D)$. Method-blind selection keeps the first case in a frozen candidate order. An independent implementation re-derived labels and causes for 224 candidate tables and found zero mismatches. Planted templates are not truth: 79–90% of realized A cases contain a cause crossing planted paths, while only 4–21% exactly match the planted cause set. Cells contain two to at least four causes, of size one to five; hence we score realized cause-set equality.

Reproducibility and cost. Model revisions, tokenizer chat templates, case order, replacements, and greedy decoding (at most 16 new tokens) are pinned. Each replay stores mask ID, raw/normalized answer, prompt SHA-256, revision, tokens, and class size. The evaluator verifies sufficiency and member-necessity against the exhaustive table; invalid subsets receive no credit. Stochastic baselines use one fixed 10-seed panel.

4.3. Attribution and accounting results

LOO has a sharp empirical boundary. Across completed registry cells, LOO obtains U ACEM 1.00 and A ACEM 0, exactly matching Proposition 1. The standard enumerator and RAG adaptation obtain A ACEM 1.00 in the completed fair cells. Hierarchical deletion and KernelSHAP also return at most one subset and therefore have A ACEM 0 under the all-causes endpoint; this is an endpoint mismatch, not evidence that their single returned subset is invalid. On monotone A, every Adapt and StdEnum return is a member of the

exhaustive $\mathcal{M}(v)$ (cause precision 1.00; invalid-return rate 0 in all four cells).

Correct caching collapses the claimed advantage. In registry A cases, the complete 256-mask tables contain only 15.7–19.8 distinct rendered prompts on average (6.1–7.7%). Under the original mask-count budget, the Qwen adaptation exceeded repeated deletion by 20.5/20.2 ACEM points on Hotpot/NQ; under unique-prompt accounting the gaps shrink to 9.5/9.6 points, below the preregistered 10-point criterion. Random-mask ACEM changes from .00/.00 to 1.00/.920 because equivalent masks no longer consume fictitious generator calls. With unique-prompt accounting, Mistral A ACEM is 1.00 for the adaptation, .913/.931 for repeated deletion, and .981/.983 for random masks on Hotpot/NQ. The standard enumerator matches 1.00 while using fewer prompts in every cell. These results reject the preregistered method-superiority criterion and show that this A slice is close to saturated. Source-level paired bootstraps give Adapt–Repeated gains (95% intervals) of .095 [.060,.136], .096 [.053,.149], .087 [.056,.122], and .069 [.042,.096] in Table 1 order; every point estimate is below the prespecified .10 margin. Adapt–StdEnum ACEM is exactly zero in every cell (Holm-adjusted exact $p = 1$). StdEnum saves .41–1.33 prompts on average; the paired 95% interval excludes zero in three cells and is [−.51, 1.17] in NQ/Qwen. Ten stochastic seeds are averaged within each source. The frozen news family replicates A ACEM 1.00 for both enumerators in all four cells ($n = 38$ –53); repeated deletion is .898–.968 and random masks .943–.976. NQ/Mistral has 38 rather than 40 sources and is descriptive. In a post-hoc $n = 10$ audit, only 1.7–2.1% of 1,024 masks are distinct prompts and both enumerators remain at 1.00. The all-active control instead has $|Q| = 256$, but yields only 6/5/3/4 A sources and shifts many candidates to N; it is a no-collapse stress test, not an accuracy benchmark.

4.4. Safety-audit and generalization results

Residual auditing is diagnostic, not confirmatory. Standard-Enum already observes violations in .267–.448 of raw N cases. Prompt-uniform residual Audit raises ten-seed mean recall to .922–.980. Averaging model outcomes within each dataset/source cluster gives .952 over 130 source clusters (222 model-cases; bootstrap 95% CI [.930,.972]) while retaining U/A ACEM 1.00. Witness soundness implies no false flags, confirmed on all 429 monotone U/A model-cases. Its paired gain over the identical enumeration prefix is .563 [.495,.630]. On detected N cases, the witness appears after 4.05 residual prompt queries on average. Figure 3 shows the budget tradeoff: at 12,16,24,32,48 prompts, mean A ACEM is .545,.967,.991,.995,1.00 and mean N recall is

Table 1. A-stratum ACEM / mean unique prompts at budget 48; n_A is in parentheses. Repeated and Random average ten seeds per source. StdEnum is the task-matched standard enumerator; Adapt is its LOO-fast-path adaptation; Audit adds prompt-uniform residual auditing and averages ten seeds.

Data	Model	LOO	Repeated	Random	StdEnum	Adapt	Audit
Hotpot (58)	Qwen	.00 (5.9)	.905 (13.0)	1.00 (15.7)	1.00 (12.4)	1.00 (13.0)	1.00 (15.7)
NQ (49)	Qwen	.00 (6.1)	.904 (14.2)	.920 (18.4)	1.00 (14.2)	1.00 (14.6)	1.00 (18.4)
Hotpot (54)	Mistral	.00 (5.9)	.913 (13.1)	.981 (15.4)	1.00 (11.9)	1.00 (12.9)	1.00 (15.4)
NQ (52)	Mistral	.00 (5.9)	.931 (13.4)	.983 (15.8)	1.00 (12.8)	1.00 (14.1)	1.00 (15.8)

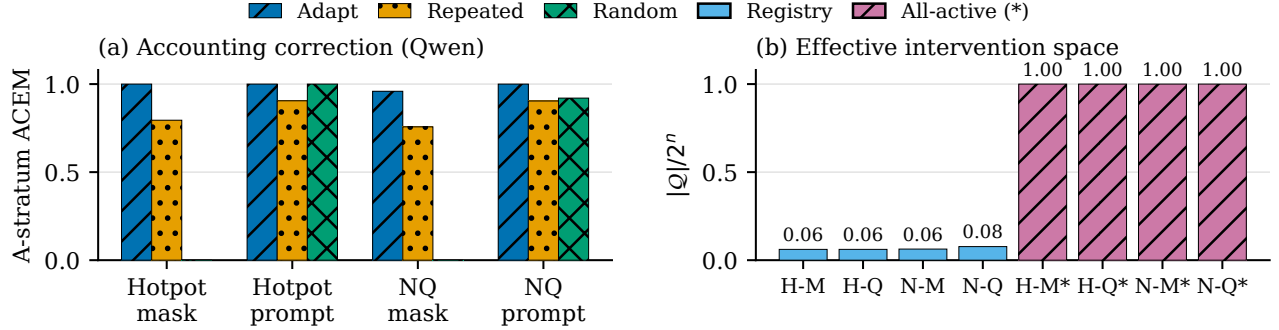


Fig. 2. The accounting audit. (a) A fixed 48-mask budget was wasted on prompt-equivalent masks; charging unique rendered prompts sharply reduces the apparent adaptation gap and rescues random masks. (b) Registry prompt spaces collapse to a small fraction of the Boolean lattice, whereas the all-active control has no literal prompt equivalence. Q/M denote Qwen/Mistral.

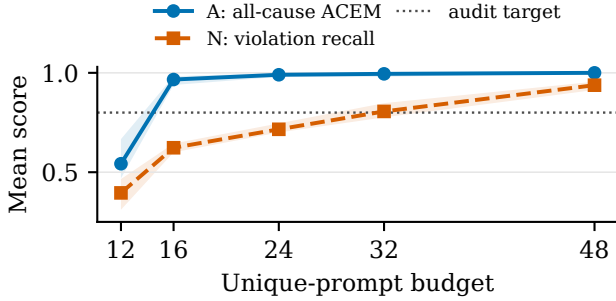


Fig. 3. Post-hoc residual-audit budget frontier on four registry cells. Lines are cell means; bands are cell minima–maxima. Cause enumeration is never preempted, while residual budget improves violation recall.

.421,.635,.734,.841,.949.

The wrapper was designed after inspecting the frozen comparison, so these are explicitly post-hoc results. We freeze it for a post-hoc transfer check on the separate news-wording family: N recall is .950 [.929,.968] over 123 clusters (207 model-cases), and A ACEM remains 1.00. The all-active control, where $|Q| = 256$, falls to .814 [.753,.869] over 75 clusters (142 model-cases); this negative stress result prevents a near-certification interpretation. Ablations also show that strategy novelty is limited: deterministic upward, downward, and cover audits obtain .797, .667, and .935 on registry N, while a randomized cover order gives .948. The robust benefit comes primarily from protecting enumeration and spending the residual physical-call budget, not from a uniquely superior boundary heuristic.

Offline replay of 1,000 uniform permutations per N case gives

fixed-prefix recall .920872 and sequential recall .920959 over 571 model-cases (138 source clusters), a difference of .000087 [−.000030,.000495], with no dynamic-only detection. This supports the static approximation only on this corpus.

We froze nearest ordering before a two-dataset/two-model hold-out. Method-blind selection retained 67/64 N cases in MSMARCO but only 15/16 in NQ, below the 30-per-cell floor. Descriptively, recall was .835 [.781,.886], with gains over uniform of .038 [.004,.072] (below the .05 margin) and shuffled cover of .028 [−.009,.064]; one cell was negative. A separately preregistered feasibility pilot on source-disjoint SQuAD and WebQuestions data [32, 33] also stopped before method evaluation: N yields 14/13/19/14 all missed its 20-source floor; no final replay or retuning followed. These blind failures accord with Proposition 3: geometry may help under a validated distribution but cannot improve the adversarial static guarantee. All methods make zero unconditional completeness claims on N; by Proposition 2, an unviolated trace is inconclusive.

5. LIMITATIONS AND CONCLUSION

The deployment rule is assumption-sensitive: validate D and $\emptyset$; under defensible monotonicity, use $v(B)$ to distinguish a unique cause from alternatives, then enumerate. Otherwise only a reversal is decisive and a clean partial trace abstains. Uniform class auditing is a small- n method, and cost must separate masks, unique prompts, and tokens. Results are local to constructed contexts, fixed replacements, exact match, and two 7B models; they do not estimate natural U/A/N prevalence. Nevertheless, they support a defensible practice: protect all-cause search, audit only with residual budget, and never promote an unviolated trace to a certificate. **AI use.** Codex assisted code and editing; authors verified all claims.

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